

SBI 811

Computer Programming in Bioinformatics

Lecture 1: Linux Foundations for Bioinformatics

Central idea	Linux is an environment for organizing, inspecting, searching and processing biological data.
Lecture progression	Understand the environment -> navigate -> inspect files -> search/extract data -> combine commands.
Biological files	FASTA, FASTQ, GFF and tabular count data.

1. Why Linux in bioinformatics?

BIOINFORMATICS DATA

Biological data may include DNA, RNA, proteins, whole genomes, sequencing reads, annotations, alignments and gene-expression measurements.

Some files contain millions of records and may occupy several gigabytes.

WHY THE COMMAND LINE?

Manual work is reasonable for a few files; it becomes inefficient when the same operation must be repeated across many samples.

Commands can be repeated, saved, modified, automated and transferred to another computer.

```
sample01_R1.fastq.gz
sample01_R2.fastq.gz
sample02_R1.fastq.gz
sample02_R2.fastq.gz
```

KEY IDEA

Small or repetitive tasks become scalable when the instructions themselves can be reused.

WORKING SPACE

2. Linux, terminal and shell

LINUX

Linux is the operating environment in which many bioinformatics tools run.

It is widely used on personal computers, servers and high-performance computing systems.

TERMINAL

The terminal is the interface where commands are typed and output is displayed.

SHELL

The shell interprets commands. Bash is one commonly used shell.

User -> Terminal -> Shell -> Linux

```
$ echo $SHELL
$ whoami
```

WINDOWS AND WSL

Ubuntu may run through WSL on a Windows computer, but the Linux commands used here remain the same.

WORKING SPACE

3. Structure of a Linux command

GENERAL FORM

command **option** **argument**

```
ls -l data
```

INTERPRETATION

ls = command

-l = option

data = argument

```
head -n 5 genes.fasta
```

INTERPRETATION

head = command

-n 5 = option

genes.fasta = argument

WORKING SPACE

4. The Linux filesystem

HIERARCHY

Linux files are organized in a hierarchy.

The highest level is the root directory: */*.

```
/
|-- home
|  |-- student
|  |-- SBI811
|  |-- project
|  |-- data
|  |-- reference
|  |-- results
```

HOME DIRECTORY

~ represents the current user's home directory.

CURRENT WORKING DIRECTORY

At any moment, the shell treats one directory as the current location.

pwd

WORKING SPACE

[illegible]

5. Paths: absolute and relative

ABSOLUTE PATH

An absolute path begins at the root directory and gives the complete location.

```
/home/student/project/reference/genome.fasta
```

RELATIVE PATH

A relative path is interpreted from the current working directory.

```
reference/genome.fasta
```

SPECIAL SYMBOLS

/ root directory

~ home directory

. current directory

.. parent directory

```
cd data/raw
pwd
cd ../..
cd reference
cd ../results
```

WORKING SPACE

6. Basic file and directory operations

CREATE

`mkdir` creates directories.

`touch` can create an empty file.

```
mkdir data reference results
touch notes.txt
```

COPY, MOVE AND RENAME

`cp` creates a copy.

`mv` moves a file or changes its name.

```
cp notes.txt notes_backup.txt
mv notes_backup.txt results/
mv notes.txt README.txt
```

REMOVE

`rm` removes files. Command-line deletion normally does not use a recycle bin.

```
pwd
ls
rm results/notes_backup.txt
```

WORKING SPACE

7. Biological file formats

FASTA

Sequence data: DNA, RNA or protein.

FASTQ

Sequencing reads plus quality information.

GFF / GTF

Genome features and coordinates.

SAM / BAM

Sequence alignments.

TSV / CSV

Tabular results such as counts.

```
>gene1
ATGCGTACGT

@read1
ATGCGTACGT
+
IIIIIIIIII
```

WORKING SPACE

8. Organizing a bioinformatics project

```
rna_seq_project/  
|-- data/  
| |-- raw/  
| `-- processed/  
|-- reference/  
|-- scripts/  
|-- results/  
|-- logs/  
`-- README.md
```

WHY THIS STRUCTURE?

raw: original experimental data

processed: derived data

reference: genome and annotation files

scripts: analysis commands/programs

results: important outputs

logs: execution messages

```
mkdir -p SBI811/week1_project/{data/raw,data/proce  
ssed,reference,scripts,results,logs}
```

WORKING SPACE

9. FASTA: searching with grep

BIOLOGICAL QUESTIONS

Which records are annotated as transporters?

How many sequence records are present?

```
head reference/genes.fasta
```

```
grep 'transporter' reference/genes.fasta
```

```
grep '^>' reference/genes.fasta
```

```
grep -c '^>' reference/genes.fasta
```

MEANING

grep searches text.

^ means the beginning of a line.

-c counts matching lines.

WORKING SPACE

10. FASTQ: using file structure

ONE FASTQ RECORD

Line 1: read identifier

Line 2: nucleotide sequence

Line 3: separator

Line 4: quality string

```
@read1
ATGCGTACGT
+
IIIIIIIIII
```

```
head -n 4 data/raw/sample01_R1.fastq
head -n 8 data/raw/sample01_R1.fastq
wc -l data/raw/sample01_R1.fastq
```

CALCULATION

If each record occupies four lines:

number of reads = number of lines / 4

```
expr $(wc -l < data/raw/sample01_R1.fastq) / 4
```

WORKING SPACE

11. GFF: structured biological fields

IMPORTANT GFF COLUMNS
3 feature type
4 start
5 end
7 strand
9 attributes
chr1 RefSeq gene 100 500 . + . ID=geneA
grep -v '^#' reference/annotation.gff
awk '{print \$3}' reference/annotation.gff
awk '\$3=="gene"' reference/annotation.gff
awk '\$3=="gene" {print \$1,\$4,\$5,\$7}' reference/annotation.gff

[illegible]

12. GFF: calculating gene length

GENOMIC COORDINATES

For inclusive coordinates:

length = end - start + 1

500 - 100 + 1 = 401

USING GFF COLUMNS

Start = column 4

End = column 5

Attributes = column 9

```
awk '$3=="gene" {print $9, $5-$4+1}'  
reference/annotation.gff
```

EXTENSION

The same command can print additional fields such as chromosome or strand.

WORKING SPACE

13. Count tables: cut, awk and sort

BIOLOGICAL QUESTIONS

Which genes have more than 100 counts in sample01?

Which gene has the highest count?

```
cut -f1 results/gene_counts.tsv
```

```
cut -f1,2 results/gene_counts.tsv
```

```
awk 'NR>1 && $2>100' results/gene_counts.tsv
```

```
tail -n +2 results/gene_counts.tsv | sort -k2,2nr
```

MEANING

-f selects fields.

\$2 means field 2.

NR>1 skips the header.

sort -k2,2nr sorts column 2 numerically from high to low.

WORKING SPACE

14. Pipes and redirection

PIPE

The pipe symbol | sends the output of one command to the next.

```
grep -v '^#' reference/annotation.gff |  
awk '{print $3}' |  
sort |  
uniq -c |  
sort -nr
```

REDIRECTION

The symbol > writes command output to a file.

```
grep '^>' reference/genes.fasta >  
results/fasta_headers.txt
```

PIPELINE LOGIC

filter -> extract -> sort -> count -> sort again

A long pipeline is a sequence of small operations.

WORKING SPACE

15. End-of-lesson challenge

DATASET

bacteria.fasta

bacteria.gff

sample_R1.fastq

sample_R2.fastq

counts.tsv

QUESTIONS

How many FASTA records are present?

How many gene features occur in the GFF file?

Which feature types occur, and how many of each?

Which genes have sample01 counts greater than 100?

Which gene has the highest count?

How can one result be saved into a new file?

CORE WORKFLOW

**Navigate -> Inspect -> Search -> Extract -> Summarize
-> Save**

```
pwd ls head grep awk cut sort uniq | >
```

WORKING SPACE